

LLM-Assisted Information Provision for Urban Atmospheric Environmental Quality Based on Spatially Aware Machine Learning Data Fusion

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Abstract The quality of the urban atmospheric environment is mainly influenced by air pollution, meteorology, traffic, land use, urban morphology and urban green spaces. Although relevant machine learning (ML) and deterministic modelling systems can provide detailed estimates and forecasts, their outputs are still difficult to translate into actionable information for decision makers, professionals, citizens and vulnerable groups. The central message of this contribution is that urban air-quality forecasting should be treated as a traceable environmental-information workflow: ML is used to fuse heterogeneous spatial evidence and provide forecasts, while large language models (LLMs) are used to produce governed, audience-specific information products. The framework combines reference observations, heterogeneous low-cost sensor data, gridded model outputs, satellite products, road-network information and OGC-compliant three-dimensional city descriptors. The methodological core consists of data access, data-warehouse or federated integration policies, QA/QC, metadata-based harmonisation, imputation, feature engineering, explainability-aware feature selection, ML model selection, ensemble forecasting and fusion, XAI evidence generation and LLM-assisted reporting. Feature prioritisation is formulated as an optimisation problem involving SHAP-based explanatory values, while local explanations such as LIME are included in the evidence package supplied to the LLM. The final system is positioned as an implementation-oriented specification for environmental information systems with data connectors, metadata catalogues, semantic harmonisation, feature stores, model registries, XAI evidence stores, curated knowledge bases and controlled dissemination channels.

Keywords air quality, environmental informatics, machine learning, low-cost sensors, data fusion, explainable AI, large language models, environmental information systems

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1 Introduction

Urban air quality-related information is increasingly produced by heterogeneous systems that combine reference stations, low-cost sensors, deterministic and data-driven air quality models, meteorological forecasts, satellite products and urban geospatial data, while involving a multiplicity of related informatics infrastructure. These systems can provide dense spatial and temporal coverage, but they also create an environmental informatics problem: data must be collected from diverse infrastructures, checked, harmonised, documented, fused, interpreted, and communicated in forms that different users can understand. The challenge is therefore not only to estimate pollutant concentrations, but also to transform uncertain, multi-source environmental data into structured, understandable and trustworthy information products.

Earlier work related to the field has underlined various challenges and limitations. Thus, work on urban air quality management systems (UAQMSs), emphasised that simplification, aggregation and operational decision support introduce deviations that must be understood and controlled [2]. More recent work on low-cost air quality sensor networks (LCAQSNs) has shown that machine learning (ML) calibration and modelling can support spatially distributed monitoring, especially when local information, reference collocation and network-level modelling are combined [1, 3]. At the same time, explainable artificial intelligence is becoming important especially when ML methods are employed, as public health and environmental decisions cannot rely only on so called black box predictions. In air quality modelling, SHAP and LIME have been shown to provide complementary global and local explanations, while SHAP-based feature reduction can reduce model complexity with limited loss of predictive performance [4, 5, 6].

Service-oriented, ontology-based and multimodal environmental information studies provide useful background for the present ML/XAI/LLM framework, especially regarding user relevance, semantic integration and personalised communication [17, 18, 19]. These works are discussed more explicitly in Section 5, where the benefits of the LLM-assisted communication layer are positioned against earlier approaches to user-tailored air-quality information.

This paper presents an environmental informatics framework for ML-powered and LLM-assisted information provision concerning the quality of the urban atmosphere. The aim is not to replace deterministic modelling, monitoring infrastructure or expert interpretation, but to create a traceable software-mediated workflow that converts heterogeneous environmental data into validated, audience-specific information.

The main message is that ML and LLMs serve different and complementary roles. ML supports the fusion of heterogeneous spatial evidence and the production of estimates, forecasts, uncertainty descriptors and explanations. The LLM layer acts as a controlled information interface that receives structured evidence and transforms it into technical reports, public advisories or audience-specific messages. In this sense, the contribution should be considered an implementation-oriented specification of an environmental information system and not as a report on a testable software demonstrator. The mathematical notation is used as a concise way to identify data objects,

transformations, functions and operations of various types and information products; it is not introduced for theorem proving.

The proposed system acts as an "integrator" of environmental modelling and environmental informatics. From the modelling point of view, it estimates or forecasts atmospheric environment states at an urban domain. From the informatics point of view, it manages heterogeneous data streams, semantic mappings, quality flags, records, models and characteristics, explanation objects and communication products. This distinction is important because operational environmental information services often fail, not because an individual model or software component is unavailable, but because the related information workflow is fragmented. Data arrive at different spatial characteristics and temporal resolutions; data curation is not always well structured; model outputs are not always accompanied by performance indices; and public-facing communication may be disconnected from the evidence that produced it. The framework below therefore treats each forecast as part of a traceable information object rather than as an isolated output of a sophisticated computational procedure.

At the end of the introductory section, Table 1 gives a short reading guide to the main symbols and abbreviations used in the framework to support the introduction of the problem first and to provide the notation needed for the formal sections that follow.

Table 1 Compact list of the main mathematical symbols and abbreviations.

Symbol	Meaning
Ω	Urban spatial domain, $\Omega \subset \mathbb{R}^2$ or $\Omega \subset \mathbb{R}^3$
$\mathbf{s}, t, h$	Location, time and forecast horizon
$Y, \hat{Y}$	True and estimated (atmospheric environmental) state
$\mathcal{D}$	Heterogeneous environmental data layer
$\mathcal{D}_{\text{OGC}}$	OGC/CityGML-compliant semantic urban-data layer
Π	Transfer operator from native support to common modelling support
$\mathbf{d}(\mathbf{s}, t)$	Harmonised multi-source input vector
$\mathbf{z}_i(t)$	Engineered feature vector at node i and time t
$\mathcal{F}, F_w$	Candidate feature set and candidate feature w
$\mathcal{A}_F$	Selected feature subset
τ_w, η_w, ψ_w	Pheromone, heuristic and SHAP-based importance of feature F_w
J_{XAI}	Explainability-aware feature-selection objective
$\mathcal{O}_d$	Daily graphical and tabular output package
$\mathcal{X}_d$	XAI-derived evidence layer
$\mathcal{L}_\theta^*$	Grounded, instructed or curated LLM
$\mathcal{M}_d$	Metadata associated with the daily evidence package
$\mathcal{K}_{\text{text}}$	Curated textual knowledge base for LLM grounding
$\mathcal{R}_{\text{domain}}$	Domain rules, thresholds and reporting constraints
AOD	Aerosol Optical Depth
CTM	Chemical Transport Model
ETL	Extract, Transform, and Load (a data integration process)
IoT	Internet of Things
LST	Land Surface Temperature
LIME	Local Interpretable Model-agnostic Explanations, local surrogate explanation method
MQTT	Message Queuing Telemetry Transport (a communication protocol designed for IoT devices)
NWP	Numerical Weather Prediction
SHAP	SHapley Additive exPlanations, global and local feature-contribution method

2 Multi-Source Environmental Data Representation

Let Ω denote the urban domain of interest and let $\mathbf{s} \in \Omega$ be a spatial location. The target of the system is an atmospheric-environmental state, for example pollutant concentration, exposure indicator or combined heat–pollution stress, estimated or forecasted as

$$\hat{Y}(\mathbf{s}, t + h) = F_{\text{fusion}}\{\mathbf{d}(\mathbf{s}, t)\}, \quad (1)$$

where h is the forecast horizon, F_{fusion} is an ML-powered data fusion function and $\mathbf{d}(\mathbf{s}, t)$ is a harmonised multi-source input vector.

The heterogeneous data layer is represented as

$$\mathcal{D} = \{\mathcal{D}_{\text{pt}}, \mathcal{D}_{\text{pix}}, \mathcal{D}_{\text{grid}}, \mathcal{D}_{\text{line}}, \mathcal{D}_{\text{OGC}}, \mathcal{D}_{\text{ctx}}\}. \quad (2)$$

Here point observations $\mathcal{D}_{\text{pt}}$ contain reference and low-cost sensor observations; $\mathcal{D}_{\text{pix}}$ contains satellite products such as atmospheric optical depth (AOD) or vegetation indices; $\mathcal{D}_{\text{grid}}$ contains gridded deterministic air quality, meteorological or reanalysis fields; $\mathcal{D}_{\text{line}}$ contains road-network and traffic information; $\mathcal{D}_{\text{OGC}}$ contains OGC-compliant urban information; and $\mathcal{D}_{\text{ctx}}$ contains additional contextual information, such as population, activity or vulnerability indicators.

Each data type has its own spatial support. Point observations are tied to monitoring sites, satellite products to pixels, deterministic model results to grid cells, transport information to line geometries and urban information to semantic objects. These data are mapped to a common input vector, by transfer operators:

$$\mathbf{d}(\mathbf{s}, t) = [\Pi_{\text{pt}}, \Pi_{\text{pix}}, \Pi_{\text{grid}}, \Pi_{\text{line}}, \Pi_{\text{OGC}}, \Pi_{\text{ctx}}] (\mathcal{D}; \mathbf{s}, t). \quad (3)$$

The operators may involve interpolation, spatial overlay, temporal aggregation, distance-decay kernels, neighbourhood summaries or semantic object queries.

Table 2 Main environmental data supports and examples of informatics operations required before ML fusion.

Support	Example data	Informatics operation
Point	Reference and low-cost sensors	calibration, QA flags, temporal alignment
Pixel	Satellite AOD, LST, vegetation indices	footprint overlay, cloud masking, resampling
Grid	CTM, NWP or reanalysis fields	interpolation, bias correction, uncertainty metadata
Line	Roads and traffic links	distance kernels, emission proxies, network queries
Object	Buildings, vegetation, land use	semantic mapping, morphology descriptors
Context	population or vulnerability data	aggregation, privacy-aware spatial joining

Table 2 summarises the data supports that are relevant to the framework. It is evident that the information system must preserve the native support and meaning of each source, before transforming it to the common modelling support. This is a central environmental informatics requirement, because a point observation, a satellite-pixel average and a deterministic-model grid-cell value do not carry the same spatial meaning.

The urban semantic component is particularly important. CityGML and related OGC-representations can address buildings, roads, vegetation, terrain, land use and other city objects using 3D geometry, topology, semantics and attributes. This allows the framework to move beyond ad hoc geospatial covariates and towards interoperable urban feature extraction. Following ontology-mediated approaches to the integration of air quality models and 3D city models [7, 8], the OGC layer is expressed as

$$\mathcal{D}_{\text{OGC}} = \{\mathcal{D}_{\text{CityGML}}, \mathcal{D}_{\text{road}}, \mathcal{D}_{\text{landuse}}, \mathcal{D}_{\text{vegetation}}, \mathcal{D}_{\text{terrain}}\}, \quad (4)$$

The transformation from semantic urban objects to ML-ready variables is denoted by

$$\mathbf{u}_{\text{OGC}}(\mathbf{s}, t) = \Phi_{\text{OGC}}(\mathcal{D}_{\text{OGC}}, \mathbf{s}, t). \quad (5)$$

The resulting descriptors may include building height, street width, street-canyon aspect ratio, sky-view factor, vegetation availability, shading potential, land-use class, road density and local morphological density. Thus, urban morphology is not treated as a background map only, but as a structured explanatory layer supporting both modelling and result interpretation. More elaborate descriptors may include orientation relative to wind direction, sky-view factor, façade continuity, tree-cover fraction, surface material and local roughness. In an environmental informatics implementation, these variables should be stored with their source object identifiers and derivation rules so that they remain traceable and reusable.

3 Environmental Informatics System Architecture

The proposed framework can be implemented as an environmental informatics software architecture, to cope with operational air quality information service requirements: the need to manage heterogeneous environmental observations, geospatial data, model outputs, metadata, uncertainty information, explanation objects and audience-specific communication products. Table 3 reports on examples of environmental informatics system components which may be used, in association with the main methodological layers.

Table 3 Examples of environmental informatics components.

Layer	Informatics function	Example components
Data access	Ingestion and interoperability	APIs, MQTT, REST, OGC services
Storage	Environmental data management	Time-series DB, PostGIS, raster store
Metadata	Traceability and provenance	Catalogue, quality flags, data lineage
Semantics	Harmonised interpretation	CityGML, vocabularies, ontology mappings
Processing	QA/QC and features	Validation rules, imputation, feature store
ML/XAI	Modelling and explanation	Model registry, ensembles, SHAP, LIME
LLM	Information provision	RAG, templates, domain rules, validation
Delivery	User-facing services	Web GIS, APIs, alerts, PDF reports

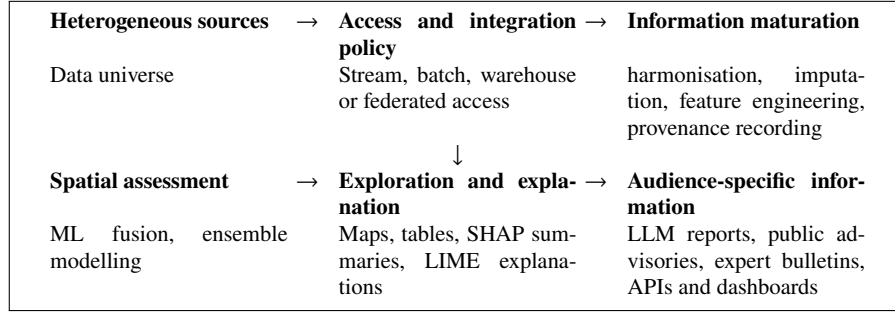


Fig. 1 Flow-oriented overview of data fusion, information maturation, spatial assessment and LLM-assisted information provision.

Figure 1 provides a flow-oriented overview of the information maturation chain. It shows where data fusion, information exploration, spatial assessment, XAI and LLM-assisted reporting enter the workflow.

The architecture also has to define an integration policy. A fully centralised data warehouse is useful for validated observations, stable feature definitions, model outputs and evidence packages because it supports reproducibility and cross-source queries [14]. However, a complete physical integration of every source is not always necessary or desirable. Some sources may remain distributed and be accessed through APIs, OGC services or query interfaces, especially when trends can be monitored without copying all raw data or when ownership, latency and update policies differ [15]. The proposed framework therefore allows a mixed architecture: high-value validated records and model-ready features can be loaded into controlled stores, while external products can be accessed in a federated manner and materialised only when needed.

Different streams also require different integration policies. High-frequency sensor data require streaming ingestion, fast QA/QC and explicit quality flags; satellite and deterministic-model products usually require batch ingestion, resampling and version metadata; CityGML and road-network layers require semantic and spatial object mapping; and vulnerability or population layers require careful aggregation and privacy-aware joining. The integration policy should therefore encode latency, reliability, spatial support, temporal resolution, ownership and uncertainty before a source is used by the ML or LLM layers.

3.1 Data Access and IoT Connectivity

The data-access layer is responsible for exposing all relevant environmental data sources to the system. These sources may include reference monitoring stations, low-cost sensor networks, deterministic air quality and meteorological model outputs, satellite products, road network information, land use data and OGC/CityGML urban morphology layers. In an operational low-cost sensor network, the field-deployed nodes may communicate through LoRaWAN gateways and an application server such as The Things Network

or The Things Stack. This layer is particularly suitable for distributed environmental monitoring, where sensor nodes must operate with low power consumption and transmit compact raw-data messages over long distances.

In this implementation pattern, the operational flow from field device to data pipeline can be written as:

$$\text{LoRaWAN sensor node} \rightarrow \text{The Things Network} \rightarrow \text{MQTT uplink} \rightarrow \text{ETL workflow.} \quad (6)$$

The Things Network or The Things Stack therefore acts as an IoT connectivity and application-integration layer. It receives sensor uplinks, manages device identity and application routing, and exposes translated or raw messages to downstream software services, for example through MQTT subscriptions.

This layer should not be confused with the scientific data-fusion layer. Its role is to provide reliable, documented and timely access to observations and external data streams. The scientific interpretation of the observations is performed later, after harmonisation, QA/QC and feature engineering.

3.2 Node-RED as ETL and Workflow-Orchestration Layer

The ETL layer connects the raw data-access interfaces with the storage and processing components. In a proposed implementation, Node-RED can be used as a lightweight, low-code orchestration environment for environmental data flows. Its role is to implement streaming and batch ETL procedures using explicit, auditable workflows.

For streaming data, Node-RED can subscribe to MQTT topics exposed by The Things Network or other IoT brokers. Each incoming message can then undergo raw-data translation, converted to physical units, enriched with metadata and routed to the appropriate storage service. For batch data, Node-RED flows or scheduled services can download satellite products, meteorological forecasts, deterministic model outputs or urban geospatial datasets at predefined intervals.

The ETL process can be summarised as:

$$\mathcal{D}^{\text{raw}} \xrightarrow{\text{Node-RED}} \mathcal{D}^{\text{etl}}, \quad (7)$$

where $\mathcal{D}^{\text{raw}}$ denotes incoming raw or semi-structured data and $\mathcal{D}^{\text{etl}}$ denotes raw-data-translated, temporally stamped, unit-harmonised and metadata-enriched records.

3.3 Ingestion and Storage

The ingestion and storage layer persists the outputs of the ETL workflows and makes them available to the downstream QA/QC, feature engineering, ML/XAI and LLM modules. A mixed storage strategy is preferable because the system handles several types

of environmental information: high-frequency sensor observations, spatial features, raster products, model outputs, metadata, explanation objects and generated reports.

MongoDB can be used as a document-oriented storage component for heterogeneous sensor messages, JSON-like observation records, metadata-rich model outputs and LLM-ready evidence packages. Its document model is suitable for records whose structure may vary between sensor types, pollutant species, calibration versions or reporting products. Time-series collections can support efficient storage and querying of timestamped observations, while geospatial indexes can support location-based retrieval of sensor nodes, stations or spatially referenced information products.

A generic environmental observation document may contain:

$\text{record} = \{\text{station_id}, \text{location}, t, \text{variables}, \text{units}, \text{quality flags}, \text{processing version}\}.$

In parallel, PostGIS or equivalent spatial databases may be used for complex geometries, road-network layers and urban-planning datasets, while raster stores or object storage may be used for satellite products and gridded deterministic model outputs. The architecture is therefore intentionally hybrid: MongoDB supports flexible document and time-series storage, while geospatial and raster databases support specialised spatial operations.

3.4 Semantic Harmonisation, Metadata and Provenance

After ingestion, the system must harmonise data semantics, spatial support, temporal resolution and metadata. This is a core environmental informatics function. Sensor observations, satellite pixels, deterministic model grids, road segments and CityGML urban objects do not share the same data model. They must therefore be mapped into a common information representation before they can be used consistently by the ML and LLM layers.

The harmonised environmental data object can be represented as:

$$\mathbf{d}(\mathbf{s}, t) = [\Pi_{\text{pt}}, \Pi_{\text{pix}}, \Pi_{\text{grid}}, \Pi_{\text{line}}, \Pi_{\text{OGC}}, \Pi_{\text{ctx}}](\mathbf{s}, t), \quad (8)$$

where the operators Π_{pt} , Π_{pix} , Π_{grid} , Π_{line} , Π_{OGC} and Π_{ctx} map point observations, satellite pixels, model grids, road-network objects, OGC/CityGML urban objects and contextual data to a common prediction support.

The OGC/CityGML layer is particularly important for semantic urban context. It allows buildings, streets, vegetation, land-use objects and other urban features to be represented not merely as geometries, but as semantically typed urban objects. These objects can be transformed into ML-ready descriptors such as building height, street-canyon aspect ratio, sky-view factor, road density, green fraction and shading potential. Metadata and provenance records should also store the source, acquisition time, processing version and quality status of each descriptor.

3.5 QA/QC, Preprocessing and Feature Store

The QA/QC and preprocessing layer converts harmonised data into trustworthy analytical input. This layer applies validation rules, range checks, instrument-specific plausibility constraints, missing-value handling, outlier detection and temporal alignment. Its output is not only a cleaned dataset, but also a set of quality flags and processing records that can be used by the modelling and reporting layers.

The cleaned and feature-engineered representation can be written as:

$$\mathbf{z}_i(t) = \mathcal{E} \left(\mathcal{Q} \left(\mathcal{D}^{\text{etl}} \right) \right), \quad (9)$$

where $\mathcal{Q}$ denotes QA/QC and preprocessing, and $\mathcal{E}$ denotes feature engineering. The feature vector $\mathbf{z}_i(t)$ may include pollutant lags, rolling statistics, meteorological variables, cyclic time encodings, road-influence descriptors, satellite-derived variables, deterministic-model outputs and OGC/CityGML-derived urban morphology indicators.

A feature store can be used to preserve the engineered features together with their origin, transformation logic, validity period and version. This is important for reproducibility: each model output and LLM-generated report should be traceable to the data snapshot, feature version and model version that produced it.

3.6 ML/XAI Engine, Model Registry and Evidence Production

The ML/XAI engine consumes the feature store and produces fused air quality estimates, forecasts, uncertainty descriptors and explanation objects. The model layer may include multiple ML-based algorithms including meta-learners, ensemble and stacking combinations. Feature selection can be performed through optimisation-assisted procedures, including ant colony or genetic algorithm search, and can be guided by SHAP-based explainability scores.

The forecast can be expressed as:

$$\hat{Y}(\mathbf{s}, t + h) = F_{\text{ML}}(\mathbf{z}(\mathbf{s}, t)), \quad (10)$$

where $\hat{Y}(\mathbf{s}, t + h)$ is the predicted atmospheric-environmental state at location $\mathbf{s}$ and forecast horizon h .

The XAI layer generates explanation products such as global SHAP rankings, local LIME explanations, feature-contribution tables and feature-interaction summaries. These explanation objects are stored together with the forecasts and become part of the evidence package used by the LLM layer. The model registry maintains model versions, training periods, hyperparameter settings, calibration records and validation metrics. This supports model governance and allows the system to identify which model and feature set generated each forecast or report.

3.7 Grounded LLM Reporting and Delivery

The evidence package provided to the LLM should be structured rather than free-form. It may include forecast statistics, exceedance indicators, uncertainty summaries, predefined tables and graphs, XAI-derived explanation objects, urban-context descriptors and curated domain rules. The LLM then acts as a controlled environmental-information service that transforms model results into audience-specific reports and messages.

The LLM input package for day d can be represented as:

$$\mathcal{I}_d = \{\widehat{\mathcal{Y}}_d, \mathcal{U}_d, \mathcal{X}_d, \mathcal{O}_d, \mathcal{K}_{\text{domain}}, \mathcal{P}_{\text{aud}}\}, \quad (11)$$

where $\widehat{\mathcal{Y}}_d$ denotes forecasts, $\mathcal{U}_d$ uncertainty information, $\mathcal{X}_d$ XAI evidence, $\mathcal{O}_d$ predefined graphical and tabular outputs, $\mathcal{K}_{\text{domain}}$ curated domain knowledge and $\mathcal{P}_{\text{aud}}$ audience profiles.

The delivery layer may include dashboards, APIs, alerts, PDF reports and machine-readable information products. Human review is included as a governance mechanism. Expert annotations and review outcomes can document whether a generated product was accepted, corrected or withheld, thereby supporting metadata improvement, model monitoring, prompt refinement and reporting-rule updates without turning the information service into an unconstrained feedback loop.

4 Explainability-Aware ML Data-Fusion Pipeline

The computational workflow is organised into four compact stages. The first stage performs data access, harmonisation, QA/QC, outlier detection and imputation. The second creates a feature store containing temporal, spatial, road-network, satellite, meteorological and CityGML-derived descriptors. The third performs explainability-aware feature selection. The fourth selects and combines ML models and produces predefined information products.

In the QA/QC stage, typical checks include range validation, rate-of-change tests via derivative calculations, instrument-status checks, meteorological plausibility rules and duplicate detection. Missing values and outliers are identified and corrected and relevant information is stored for future comparisons. The second stage transforms the harmonised data into ML-ready variables. These include lags, rolling statistics, cyclic temporal encodings, spatial-neighbourhood summaries, satellite data summaries, road-influence information, city morphology, and uncertainty descriptors. Each engineered feature is linked to its source variables and transformation history.

For node i and time t , the engineered feature vector is written as

$$\mathbf{z}_i(t) = \mathcal{E}\{\mathbf{d}(\mathbf{s}_i, t), \mathbf{d}(\mathbf{s}_i, t-1), \dots, \mathbf{u}_{\text{OGC}}(\mathbf{s}_i, t)\}, \quad (12)$$

where $\mathcal{E}$ denotes feature engineering. Candidate features include current and lagged sensor values, meteorological indicators, cyclic temporal variables, satellite summaries,

neighbourhood statistics, road-influence variables, urban morphology descriptors and uncertainty indicators.

Let $\mathcal{F} = \{F_1, \dots, F_W\}$ be the full candidate-feature set. The selected subset $\mathcal{A}_F \subseteq \mathcal{F}$ is obtained through an optimisation process. In an ant colony optimisation (ACO)-inspired formulation, each artificial ant constructs a candidate subset by moving through the feature-subset space. Feature F_w has a pheromone value $\tau_w^{(r)}$ at iteration r , a heuristic relevance score η_w and a SHAP-based explanatory score ψ_w . The probability of selecting F_w can be written as

$$P_w^{(r)} = \frac{(\tau_w^{(r)})^\alpha (\eta_w)^\beta (\psi_w)^\gamma}{\sum_{F_j \in \mathcal{F}_{\text{available}}} (\tau_j^{(r)})^\alpha (\eta_j)^\beta (\psi_j)^\gamma}, \quad (13)$$

where α , β and γ control the relative influence of optimisation memory, heuristic relevance and explainability. The SHAP score is computed as

$$\psi_w = \frac{1}{N_s} \sum_{n=1}^{N_s} |\phi_{n,w}|, \quad (14)$$

where $\phi_{n,w}$ is the SHAP value of feature F_w for sample n [5]. This formulation allows the feature-selection process to favour variables that are predictive and also interpretable.

Candidate feature subsets are evaluated by an explainability-aware objective function:

$$J_{\text{XAI}}(\mathcal{A}_F) = \alpha_1 E_{\text{pred}}(\mathcal{A}_F) + \alpha_2 E_{\text{sp}}(\mathcal{A}_F) + \alpha_3 U(\mathcal{A}_F) - \alpha_5 E_{\text{SHAP}}(\mathcal{A}_F). \quad (15)$$

Here E_{pred} measures predictive error, E_{sp} penalises spatial inconsistency of residuals, U represents uncertainty or calibration quality and E_{SHAP} rewards explanatory value. Pheromone trails are then updated so that features participating in high-quality subsets receive stronger reinforcement, while unused or weak features lose influence through evaporation. It should be mentioned that the same selection logic can be combined with genetic algorithms or other metaheuristics.

This optimisation is useful in air quality modelling because many environmental predictors are correlated; a purely greedy ranking may discard variables that are weak alone but useful in interaction with meteorology, urban morphology or traffic.

The modelling stage may include various ML-based algorithms. A generic learner f_l with hyperparameters θ_l produces

$$\hat{Y}_l(\mathbf{s}, t + h) = f_l(\mathbf{z}(\mathbf{s}, t; \mathcal{A}_F), \theta_l). \quad (16)$$

The final estimate may be an ensemble. Graph neural networks can exploit node connectivity derived from distance, road networks, wind direction or similarity in urban morphology [10, 11]. Gaussian-process residual fields can represent spatially structured errors and provide uncertainty-aware corrections [9]. The important informatics point is that each prediction is stored together with the data snapshot, feature subset, model version, uncertainty descriptor and explanation layer.

The model registry and feature store are therefore part of the scientific method, not merely implementation conveniences. The registry records the learner family, hyperpa-

rameters, training period, validation strategy and operational status of each model. The feature store records the definition, units, spatial support, temporal validity and source dependencies of each engineered feature. When a model is retrained, the system can compare not only predictive metrics but also the stability of the selected features and explanations. This is important for detecting problems like concept drift, sensor ageing, changes in traffic patterns or changes in the availability of urban data layers.

5 XAI- and LLM-Assisted Environmental Information Products

The modelling layer produces a daily output package O_d containing for example pre-defined tables, graphs, statistics and exceedance indicators. This package may be complemented by an XAI-derived evidence layer

$$\mathcal{X}_d = \{\Psi_d^{\text{global}}, \Lambda_d^{\text{local}}, \Gamma_d^{\text{int}}\}, \quad (17)$$

where Ψ_d^{global} contains global SHAP summaries, Λ_d^{local} contains local explanations such as LIME or instance-level feature contributions, and Γ_d^{int} contains interaction or co-dependence information. SHAP, or SHapley Additive exPlanations, assigns each feature a contribution value derived from cooperative-game-theory reasoning, thereby supporting both global driver rankings and local feature-attribution summaries [5, 4]. LIME, or Local Interpretable Model-agnostic Explanations, builds a simpler local surrogate model around one prediction in order to explain why that specific prediction was produced [6]. In this framework, SHAP is useful for identifying model-wide drivers and interactions, whereas LIME supports local explanations for individual forecasts.

This distinction is important for environmental reporting. A global SHAP summary can support statements such as “traffic-related variables and wind speed dominated the daily pollution pattern”, while a local explanation can justify why one specific location or hour was assigned a high pollution message. Interaction information can further distinguish between influencing parameters (drivers) that act independently and drivers that become relevant only under particular conditions, such as low wind speed combined with narrow street geometry or high particulate levels combined with wintertime temperature patterns.

In the current research, the LLM does not receive free-form model output. Instead, it receives a structured, auditable input package

$$\mathcal{I}_d = \{O_d, \mathcal{X}_d, \mathcal{D}_{\text{OGC}}, \mathcal{M}_d, \mathcal{R}_{\text{domain}}, \mathcal{K}_{\text{text}}, \mathcal{P}_{\text{aud}}\}, \quad (18)$$

where $\mathcal{M}_d$ is metadata, $\mathcal{R}_{\text{domain}}$ contains thresholds and reporting rules, $\mathcal{K}_{\text{text}}$ is a curated knowledge base and $\mathcal{P}_{\text{aud}}$ defines audience profiles. The grounded LLM $\mathcal{L}_\theta^*$ generates a technical or user-oriented report as

$$\mathcal{Y}_d^{\text{LLM}} = \mathcal{L}_\theta^*(\mathcal{I}_d, \mathbf{p}_g), \quad (19)$$

where $\mathbf{p}_g$ is the constrained prompt or reporting template for audience group g .

Table 4 Examples of XAI- and LLM-assisted environmental information products.

Product	Main audience	Evidence and LLM-added value
Technical report	decision makers	maps, uncertainty, SHAP drivers, metadata and concise strategic synthesis
Operational bulletin	agencies and operators	exceedances, model confidence, local explanations and suggested follow-up checks
Public advisory	citizens	simplified risk levels, plain-language explanations and practical guidance
Messages for vulnerable groups	sensitive users	location, time window, exposure, uncertainty and sensitivity-specific advice
Web-GIS layer	analysts and planners	gridded forecasts, CityGML descriptors, XAI summaries and links to supporting maps
Machine-readable API	downstream systems	JSON products, provenance, quality flags and report-generation metadata

The resulting information products may be designed for different levels of technical detail, as shown in Table 4. The important requirement is that each product is derived from the same evidence package, although the vocabulary, level of detail and recommended actions may differ.

The LLM layer addresses a specific communication gap in current air-quality information services. Earlier environmental-information and decision-support work has shown that users are often confronted with heterogeneous, partly inconsistent and difficult-to-interpret information, and that the central questions are not only which information can be trusted, but also which information is relevant to the individual user [17]. Related work on multimodal air-quality communication and ontology-based decision support further showed the need to connect environmental conditions with location, activity, user profile and health sensitivity [19, 18].

In the proposed framework, LLM-assisted reporting transforms the same validated evidence package into different communication products without changing the underlying scientific content. The LLM does not replace data fusion, QA/QC, modelling, uncertainty estimation or XAI. Instead, it converts their structured outputs into audience-specific explanations for citizens, sensitive individuals, technical operators, public-health experts and policy makers.

The role of the LLM is therefore structured and controlled information provision. For decision makers, reports can summarise model performance, expected exceedances, spatial patterns, uncertainty and dominant drivers. For environmental agencies and operators, the LLM can generate operational bulletins that distinguish between validated observations, forecasts, model confidence and suggested follow-up checks. For citizens, messages can use simpler language and focus on practical information related to everyday utility. For vulnerable groups or outdoor workers, the same evidence package can be filtered by activity profile, sensitivity, location and time window. For analysts and planners, the LLM can point to maps, CityGML-derived spatial context and XAI products that support further information exploration. Because the LLM receives SHAP/LIME information, it can explain forecasts in terms of model-supported drivers, for example

local meteorology, street-canyon morphology, reduced vegetation or meteorological stagnation.

The framework is model-agnostic with respect to the language model. A large hosted LLM, a locally deployed open model or a domain-adapted small language model (SLM) can be used, provided that it satisfies the required criteria for faithfulness, latency, cost, privacy, language coverage and controllability. An SLM may be sufficient when the reporting task is template-constrained and the evidence package is highly structured. A larger LLM may be preferable when the system must generate richer multi-audience explanations, compare several evidence objects or adapt the same evidence package to different professional and public communication styles. In the illustrative example below, Gemini 2.0 Flash is used as one possible LLM and not as a prescribed technology choice.

A practical governance rule is that the LLM should operate under a restricted reporting grammar for high-stakes messages. For example, terms such as “high”, “moderate”, “uncertain”, “expected exceedance” or “main driver” should be linked to predefined thresholds or XAI evidence. The system may also maintain an audit trail containing the prompt template, model version, retrieved knowledge items, input data snapshot and validation outcome. This turns LLM reporting into a controlled environmental-information service rather than an unconstrained text generation step.

5.1 Illustrative LLM-Generated Air-Quality Report

To demonstrate the operational role of the LLM layer, an illustrative PM10 forecast report was generated using Gemini 2.0 Flash. The report was produced for the reference air quality monitoring station of Agia Sofia, in Thessaloniki, Greece (station code GR0018A), for a one-hour-ahead forecast horizon, $t + 1$. The structured input provided to the LLM included predicted PM10 concentration statistics, model-performance indicators, diagnostic plots, model-combination information and reporting rules. The purpose of this example is not to evaluate the forecasting model itself, but to illustrate how the proposed architecture can transform upstream model outputs into an environmental-information product.

Table 5 presents the first page of the generated report as a large thumbnail, together with a compact explanation of its role in the proposed framework. The report translates numerical model outputs into a readable air-quality assessment, compares the forecast with relevant reference values, explains the hybrid stacked ensemble in plain language and introduces recommendations for both the public and air-quality authorities.

The example also demonstrates why the LLM layer should be connected to the XAI and verification components of the architecture. The generated text does more than restate the predicted concentration; it explains the hybrid stacked ensemble, interprets diagnostic plots and translates model results into actionable information. However, such interpretation must remain traceable to the underlying evidence package. In operational use, the report should therefore be treated as a governed information product grounded in model outputs, diagnostic evidence, domain knowledge and audience-specific reporting rules.

Table 5 Illustrative LLM-generated PM10 forecast report.

Generated report thumbnail

PM10 Air Quality Forecast Report

Station: GR0018A • Generated: 2024-03-01 14:30:00 • Forecast: t+1 (next hour)

PM10 Predicted Mean: 37.05 $\mu\text{g}/\text{m}^3$	Air Quality Category: Moderate	Model RMSE: 8.8 $\mu\text{g}/\text{m}^3$	R^2 : 0.7916
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1. Air Quality Assessment

The forecasted PM10 concentration for station GR0018A at the t+1 horizon is summarized with a mean of 37.05 $\mu\text{g}/\text{m}^3$, categorizing the air quality as "Moderate."

Compared to the WHO guideline (45 $\mu\text{g}/\text{m}^3$), the predicted mean of 37.05 $\mu\text{g}/\text{m}^3$ is below this guideline, suggesting the average air quality is moderate.

- WHO 24-hour mean guideline: 45 $\mu\text{g}/\text{m}^3$. The predicted mean of 37.05 $\mu\text{g}/\text{m}^3$ is below this guideline, suggesting the average air quality is moderate.
- EU daily limit value: 50 $\mu\text{g}/\text{m}^3$. The predicted mean of 37.05 $\mu\text{g}/\text{m}^3$ is also below this limit.

However, it is crucial to note the predicted range. PM10 levels are expected to vary significantly, with a minimum of 6.97 $\mu\text{g}/\text{m}^3$ and a maximum of 88.18 $\mu\text{g}/\text{m}^3$. While the mean falls within the "Moderate" category and below key guidelines, the predicted maximum value of 88.18 $\mu\text{g}/\text{m}^3$ is considerably higher than both the WHO guideline and the EU daily limit. This indicates that despite an overall moderate forecast, there is a potential for short-term exceedances, especially PM10 levels within the forecast period, which could pose risks, especially to sensitive populations.

2. Methodology Explanation

The PM10 forecast is generated using a sophisticated "Hybrid Stacked Ensemble" approach, which can be thought of as combining the strengths of multiple expert models to make a more accurate final prediction. Here is a breakdown in plain language:

- Two "Experts" (Base Models):
 - ANN Ensemble (Artificial Neural Network): This model is like a team of specialized learners, with each learner trained on data from a specific station. Their individual predictions are then averaged to form one "expert" or the PM10 level. ANNs are good at recognizing complex patterns in data.
 - Global k-NN: This is another powerful machine learning model, based on a larger, combined dataset from all stations. It excels at identifying important features and making robust predictions.
- A "Coordinator" (Meta-model): Instead of simply averaging the predictions from the two base models, a "Linear Regression" meta-model acts as a coordinator. It learns how much to "trust" or "weight" each expert's prediction (0.2089 for ANN and 0.8910 for Global k-NN) based on its historical performance, along with a small adjustment (intercept). This allows the system to optimally combine the individual predictions into a single, more reliable final forecast.

Interpretation within the proposed framework

The report is an example of an LLM-assisted environmental-information product. It was generated from a structured evidence package containing the PM10 forecast, model-performance indicators and diagnostic information. In the example, the forecast refers to station GR0018A and to a one-hour-ahead prediction horizon ($t + 1$). The generated report summarises the predicted PM10 mean (37.05 $\mu\text{g}/\text{m}^3$), assigns an air quality category (*Moderate*), reports the model RMSE (8.8 $\mu\text{g}/\text{m}^3$) and R^2 value (0.7916), and provides a plain-language interpretation of the forecast.

6 Evaluation

The framework should be evaluated at five levels. First, predictive performance can be assessed using statistical metrics like *RMSE*, *MAE*, correlation, R^2 , and regulatory metrics such as the relative expanded uncertainty. Spatial validation should include hold-out stations and temporally separated testing, because random splits can overestimate performance in autocorrelated environmental time series. Second, feature-selection performance should be evaluated by comparing the selected subset with the full feature set in terms of accuracy, uncertainty, stability and model complexity. A key criterion is whether for example ACO/SHAP-guided feature selection maintains predictive performance while improving interpretability and reducing redundancy.

Third, explanation quality should be evaluated by testing the consistency of global and local explanations, the stability of SHAP rankings across time and stations, and the agreement between explanation outputs and domain knowledge. For example, persistent high importance of traffic, PM-related variables, low wind speed or street-canyon descriptors should be physically plausible. Fourth, LLM-generated information products should be evaluated for faithfulness, unsupported claims, clarity, audience suitability and operational usefulness. A simple verification rule is that each statement about a driver, exceedance, uncertainty level or location-specific condition must be traceable to O_d , X_d , metadata or curated domain rules.

A fifth evaluation dimension concerns the information-system layer. The workflow should be tested for latency, completeness of metadata, reproducibility of outputs, robustness, and consistency of product generation under missing or delayed input streams. For operational use, it is not sufficient for the model(s) to perform well under ideal data availability; the system must also behave predictably when a sensor fails, a satellite

product is unavailable, a model file arrives late or a CityGML layer lacks an expected attribute.

The evaluation should therefore combine numerical metrics with information-quality indicators. Examples include the percentage of LLM statements supported by structured evidence, the stability of selected features across validation windows, and the frequency of human-review interventions. These indicators make the environmental informatics contribution measurable.

Finally, the evaluation should distinguish between scientific validation and service validation. Scientific validation asks whether the forecast and the explanations are "correct enough" for their intended purpose. Service validation asks whether the complete workflow can repeatedly deliver the right product to the right user at the right time, with the required metadata and review status. Both are necessary if the proposed framework is to move from theory or from a research prototype to an operational environmental information service. This transition stage has been supported by a multiplicity of research results so far, and its implementations are being tested under operational conditions.

7 Discussion

The proposed environmental informatics framework treats urban air quality prediction as an end-to-end information management problem. Data fusion, ML modelling, XAI and LLM reporting are not independent modules; they are connected through metadata, semantic harmonisation, feature stores, model registries and auditable evidence records. This matters because air quality information is often used in contexts where trust, traceability and clarity are as important as numerical accuracy.

The framework also clarifies the role of different modelling paradigms. Deterministic models contribute process-based physical consistency and regional context; sensor networks contribute local observations reflecting local conditions; satellite products contribute spatially extensive information; and ML models combine these sources in a data-adaptive way. The LLM layer should not be interpreted as a scientific model: its role is to transform validated structured evidence into language, provided that the model results, rules and verification procedures are explicit.

The CityGML/OGC component strengthens the framework by giving urban morphology a formal semantic representation, although the framework can also be applied if such data are not available for an urban area of interest. Urban morphology descriptors can improve predictive models, but they also improve explanations. An elevated predicted concentration in a narrow, poorly ventilated street canyon can be reported differently from an elevated concentration associated mainly with regional transport or meteorological stagnation.

The ACO/SHAP feature selection method offers a way to combine optimisation and explainability. Pheromone trails retain memory of successful feature subsets, while SHAP values favour features that contribute substantially and consistently to model behaviour. This reduces the risk of selecting variables only because they improve a numerical score without supporting interpretation. LIME or other local methods complement this by explaining particular forecasts or alerts.

Although the present contribution focuses on the urban atmospheric environment, the same information-system logic can be transferred to related domains that require heterogeneous data integration, spatial assessment and audience-specific communication. Disaster management, smart agriculture and smart forestry also combine observations, model outputs, geospatial context, uncertainty and decision-oriented reporting. The proposed architecture is therefore not limited to air quality, although the domain rules, evidence packages and validation criteria would have to be adapted.

Several limitations remain. The framework is methodological and must be tested in real operational deployments. The quality of LLM outputs depends on the quality of structured inputs, domain rules and verification. OGC/CityGML data may be incomplete or outdated in some cities. SHAP and LIME are post hoc explanation methods and can be unstable under correlated features or distribution shifts. Finally, institutional use requires governance procedures, human review, privacy safeguards and clear responsibility for public messages.

8 Conclusions

This paper presented an environmental informatics framework for LLM-assisted urban atmospheric-environmental information provision. The framework integrates multi-source data representation, semantic urban context, spatially aware ML data fusion, explainability-aware feature selection, XAI evidence generation and grounded LLM reporting.

The central idea is that air quality forecasts should be transformed into traceable environmental information products. The system should know which data were used, how they were harmonised, which features were selected, which model version generated the forecast, what the uncertainty was and which SHAP/LIME evidence supports the explanation. This makes it possible for LLMs to generate technical reports and audience-specific messages that are not merely fluent, but grounded in environmental data, model outputs and explicit explanation evidence.

A first implementation of the framework should therefore prioritise a small number of robust design principles: preservation of native data semantics, explicit transfer to common modelling support, reproducible feature engineering, explainability-aware model selection, storage of XAI evidence and constrained LLM generation. These principles allow the approach to remain extensible while keeping the overall approach focused on information provision.

The proposed framework is based on many years of research in urban air-quality information services, low-cost sensor networks and ML-assisted environmental assessment. Several components have already been developed and tested to a large degree in a service-provision orientation. The enrichment with an LLM layer is therefore presented as an added controlled interface for providing human-centred, quality-of-life-oriented information related to the urban atmospheric environment [13].

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